



OPEN Multi-modal federated learning with differential privacy for privacy-preserving healthcare AI

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The growing adoption of artificial intelligence in healthcare highlights the need for models that can leverage heterogeneous patient data while preserving strict privacy requirements. This paper proposes a novel multi-modal federated learning framework with differential privacy for decentralized healthcare AI. The model integrates electronic health records and ECG time-series using modality-specific encoders and a shared latent fusion network, enabling comprehensive representation learning without centralizing sensitive data. Differential privacy is incorporated into local updates to provide formal guarantees against information leakage in federated aggregation. Extensive experiments on real-world healthcare datasets show that the proposed method achieves 94.12% accuracy, 93.64% precision, 93.21% recall, 93.42% F1-score, and 95.03% AUC, outperforming centralized, single-modality, and non-private baselines. The framework also converges 32.4% faster than single-modality federated learning, reaching 90% accuracy in 35 rounds. An ablation study confirms the contribution of multi-modal fusion and class balancing, while client variance analysis shows the lowest performance deviation ($\pm 1.2\%$) under heterogeneous distributions. These results indicate that combining federated optimization, differential privacy, and multi-modal learning provides an effective framework for privacy-preserving clinical AI, with potential for deployment in distributed healthcare settings.

Keywords Federated learning, Multi-modal learning, Privacy-preserving AI, Healthcare AI, Electronic health records

Artificial intelligence (AI) has emerged as a transformative force in healthcare, driving advancements in disease diagnosis, risk prediction, and personalized treatment planning. With the growing availability of electronic health records (EHRs), physiological signals, and other clinical data sources, machine learning models have demonstrated their ability to uncover patterns that support timely and accurate medical decision-making^{1–3}. However, despite these advances, the use of AI in healthcare is challenged by two critical factors: the need to integrate heterogeneous patient data for comprehensive analysis and the imperative to safeguard sensitive medical information in compliance with strict privacy regulations^{4,5}. Traditional centralized training approaches require aggregating data in a single location, which raises significant legal, ethical, and security concerns, especially when handling highly sensitive patient records distributed across multiple institutions.

Within this context, federated learning (FL) has emerged as a promising paradigm that enables collaborative model training without transferring raw data outside institutional boundaries^{6,7}. By allowing each participating hospital or clinic to train models locally and only share aggregated updates, FL provides a natural fit for privacy-sensitive healthcare applications⁸. While several studies have demonstrated the effectiveness of FL for single-modality datasets such as EHRs or medical images, a key limitation of current approaches is their inability to fully exploit the complementary information available across multiple data modalities⁹. Recent research has also explored multimodal federated learning frameworks that integrate heterogeneous data sources across distributed clients. These approaches employ modality-aware fusion strategies and distributed training mechanisms to improve predictive performance in healthcare applications^{10,11}. In real-world clinical environments, patient care is informed not only by structured health records but also by physiological signals such as ECG time-series. Integrating these diverse data sources is essential for building robust and generalizable predictive models.

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Existing studies in healthcare AI have primarily examined federated learning, multi-modal learning, and differential privacy as separate research directions or through limited combinations. Many federated learning approaches are restricted to single-modality data, which limits their ability to exploit complementary information from heterogeneous clinical sources. In contrast, most multi-modal models are developed in centralized settings that assume joint access to all modalities, which is often unrealistic in distributed healthcare environments. In addition, prior studies have incorporated differential privacy into federated learning, but mainly in simpler single-modality settings, leaving its interaction with multi-modal representation learning underexplored.

In this context, the proposed framework unifies multi-modal fusion, federated optimization, and differential privacy within a single architecture. The model enables local institutions to learn from both electronic health records and ECG time-series through modality-specific encoders and a shared fusion mechanism without transferring raw data to a central server. Differential privacy is incorporated into the local update process to provide formal protection against information leakage during federated aggregation, supporting deployment in realistic distributed healthcare environments.

The contribution of this work lies in the unified integration and systematic evaluation of these components rather than the introduction of a new standalone algorithm. While prior studies have explored federated learning, multi-modal learning, and differential privacy individually or in limited combinations, the proposed framework jointly analyzes their interaction under heterogeneous data distributions, client variability, and privacy constraints. This integrated perspective provides practical insights into how these components behave together in realistic healthcare settings, highlighting improvements in performance, robustness, and communication efficiency.

Motivated by this gap, the present research addresses the problem of designing a privacy-preserving, multi-modal federated learning framework capable of jointly leveraging structured and temporal healthcare data. The primary objectives of this study are to:

- Develop a novel federated architecture that integrates electronic health records and ECG signals without requiring centralized data aggregation.
- Incorporate differential privacy into the training process to provide formal guarantees against data leakage while maintaining model utility.
- Design a modality-specific encoder and a shared fusion network to effectively capture complementary patterns across heterogeneous data sources.
- Evaluate the robustness of the proposed framework under non-identically distributed data and client heterogeneity, reflecting realistic healthcare environments.
- Demonstrate the scalability and communication efficiency of the model to ensure practical applicability in distributed clinical networks.

The remainder of this paper is organized as follows. Section “[Related work](#)” reviews the existing literature on federated learning, multi-modal representation learning, and differential privacy in healthcare. Section “[Methodology](#)” describes the proposed methodology, including the architecture, training procedure, and privacy mechanism. Section “[Results](#)” reports and analyzes the results, including comparative performance, ablation studies, and visualization of learned representations. Section “[Discussion](#)” discusses the implications, limitations, and future directions of the work. Finally, Section “[Conclusions](#)” concludes the paper by summarizing the key contributions and potential impact of the proposed framework on real-world healthcare AI applications.

Related work

Federated learning has gained significant attention as a paradigm for enabling collaborative model training without requiring direct data sharing, making it particularly attractive for sensitive domains such as healthcare¹². Traditional machine learning approaches typically rely on centralized data aggregation, which poses substantial privacy risks and often violates stringent regulations surrounding medical data. Federated learning addresses these issues by allowing multiple institutions to jointly train models while keeping patient information within local silos. Numerous studies have explored its use for electronic health record analysis, medical imaging diagnostics, and predictive modeling in hospital networks, showing its potential to facilitate large-scale clinical AI without compromising data confidentiality¹³. This paradigm has also been extensively studied in medical imaging to enable cross-institutional collaboration while preserving data locality and regulatory compliance^{14,15}. However, a common limitation of much of this work is the focus on single-modality datasets, which neglects the rich and complementary information available across different healthcare data sources.

In parallel, multi-modal representation learning has emerged as a powerful technique in medical AI by combining diverse types of patient information, such as structured electronic health records, physiological signals, and imaging data, to provide more comprehensive predictions^{16,17}. Models that integrate multiple modalities often demonstrate improved diagnostic accuracy and better capture complex clinical patterns compared to single-modality approaches. These methods leverage the interdependencies between data types to create richer feature representations that support tasks such as disease progression modeling, risk prediction, and personalized treatment planning¹⁸. Despite their effectiveness, most multimodal learning frameworks are designed for centralized data settings where all modalities are available in a single location, which is rarely feasible in real-world healthcare environments.

Another line of research relevant to privacy-preserving healthcare AI is the incorporation of differential privacy into machine learning models¹⁹. Differential privacy introduces mathematical guarantees that limit the risk of exposing individual-level information by perturbing data or model updates with calibrated noise²⁰. In the context of healthcare, this has been applied to protect electronic health records, medical images, and other sensitive datasets during model training and analysis^{21,22}. When combined with federated learning, differential

privacy provides a dual layer of protection by safeguarding both the distributed training process and the model's outputs. While promising, integrating differential privacy often introduces a trade-off between privacy strength and model performance, and many studies struggle to balance these competing demands, especially under non-identically distributed data conditions that are typical in federated settings. Recent work has further emphasized how noise injection strategies influence the trade-off between privacy guarantees and model robustness in federated learning systems²³.

Despite advances in each of these areas, there remains a noticeable gap in approaches that effectively integrate federated learning, multi-modal data fusion, and differential privacy into a single cohesive framework for healthcare applications^{24,25}. Most existing federated models are limited to a single data modality and cannot fully exploit heterogeneous information sources across clinical networks. Conversely, multi-modal models that combine different data types typically assume centralized access to all modalities, ignoring the decentralized reality of most healthcare infrastructures^{26,27}. Similarly, while differential privacy has been studied extensively in both centralized and federated contexts, its impact on complex multimodal architectures and the interactions among noise injection, communication efficiency, and predictive performance remain to be thoroughly explored. In addition, data heterogeneity across clients remains a fundamental challenge in federated healthcare systems, affecting convergence stability and consistency of model performance²⁸. These gaps highlight the need for an integrated solution capable of unifying these techniques to achieve both strong predictive performance and robust privacy protection.

The proposed model directly addresses these limitations by combining federated learning with a multimodal architecture that jointly processes electronic health records and ECG time-series data under strict differential privacy guarantees. By employing modality-specific encoders and a shared latent fusion network, the framework captures complementary patterns between structured and temporal data without requiring centralized aggregation. This design aligns with the practical constraints of distributed healthcare systems, where institutions must collaborate without sharing raw patient information. Furthermore, integrating differential privacy provides formal confidentiality guarantees, mitigating the risk of data leakage or membership inference attacks during federated aggregation. Through this combination, the proposed approach bridges the gap between existing single-modality federated models and centralized multi-modal learning methods, offering a flexible and privacy-compliant solution tailored for distributed clinical environments.

Methodology

This section presents the proposed *Multi-Modal Federated Learning with Differential Privacy* framework for privacy-preserving healthcare AI. The methodology integrates heterogeneous data modalities—tabular patient health records and ECG time-series—within a federated learning (FL) paradigm augmented with differential privacy (DP). The entire pipeline is divided into four key components: (i) federated learning architecture, (ii) multi-modal feature extraction, (iii) aggregation with differential privacy, and (iv) evaluation metrics.

Data preprocessing

The proposed framework utilizes two heterogeneous healthcare datasets: (i) the *Federated Health Records* dataset ($\mathcal{D}_H$) containing tabular patient information, and (ii) the *ECG Heartbeat Categorization* dataset ($\mathcal{D}_E$) consisting of time-series electrocardiogram signals. To ensure robust and privacy-preserving training, we employed a deep learning-oriented preprocessing pipeline that generates training, validation, and testing sets with consistent normalization and feature representation across federated clients, as presented in Fig. 1.

Data normalization and standardization

Let $\mathbf{x}_i \in \mathbb{R}^d$ denote a feature vector for the i -th patient record in $\mathcal{D}_H$. We applied z-score normalization to each feature to stabilize gradient updates across clients:

$$\tilde{\mathbf{x}}_i = \frac{\mathbf{x}_i - \boldsymbol{\mu}}{\boldsymbol{\sigma}}, \quad (1)$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ are the mean and standard deviation computed feature-wise across $\mathcal{D}_H$. This ensures each feature has zero mean and unit variance:

$$\mathbb{E}[\tilde{\mathbf{x}}] = 0, \quad \text{Var}[\tilde{\mathbf{x}}] = 1. \quad (2)$$

For the ECG dataset, each heartbeat sample is represented as a time-series $\mathbf{s}_j \in \mathbb{R}^T$ of length $T = 187$. We applied min-max scaling to the amplitude values:

$$\hat{s}_{j,t} = \frac{s_{j,t} - s_{\min}}{s_{\max} - s_{\min}}, \quad t = 1, \dots, T, \quad (3)$$

where $s_{\min}$ and $s_{\max}$ are computed per-patient to preserve local signal dynamics. This transforms the ECG signals into the range $[0, 1]$.

Dimensionality alignment and feature embedding

To enable multi-modal fusion, we embedded both datasets into a common latent space $\mathbb{R}^k$. For tabular records, a nonlinear encoder $\phi_H(\cdot)$ parameterized by a multilayer perceptron (MLP) maps $\tilde{\mathbf{x}}_i \mapsto \mathbf{z}_i \in \mathbb{R}^k$:

$$\mathbf{z}_i = \phi_H(\tilde{\mathbf{x}}_i) = \sigma(W_2 \sigma(W_1 \tilde{\mathbf{x}}_i + \mathbf{b}_1) + \mathbf{b}_2), \quad (4)$$

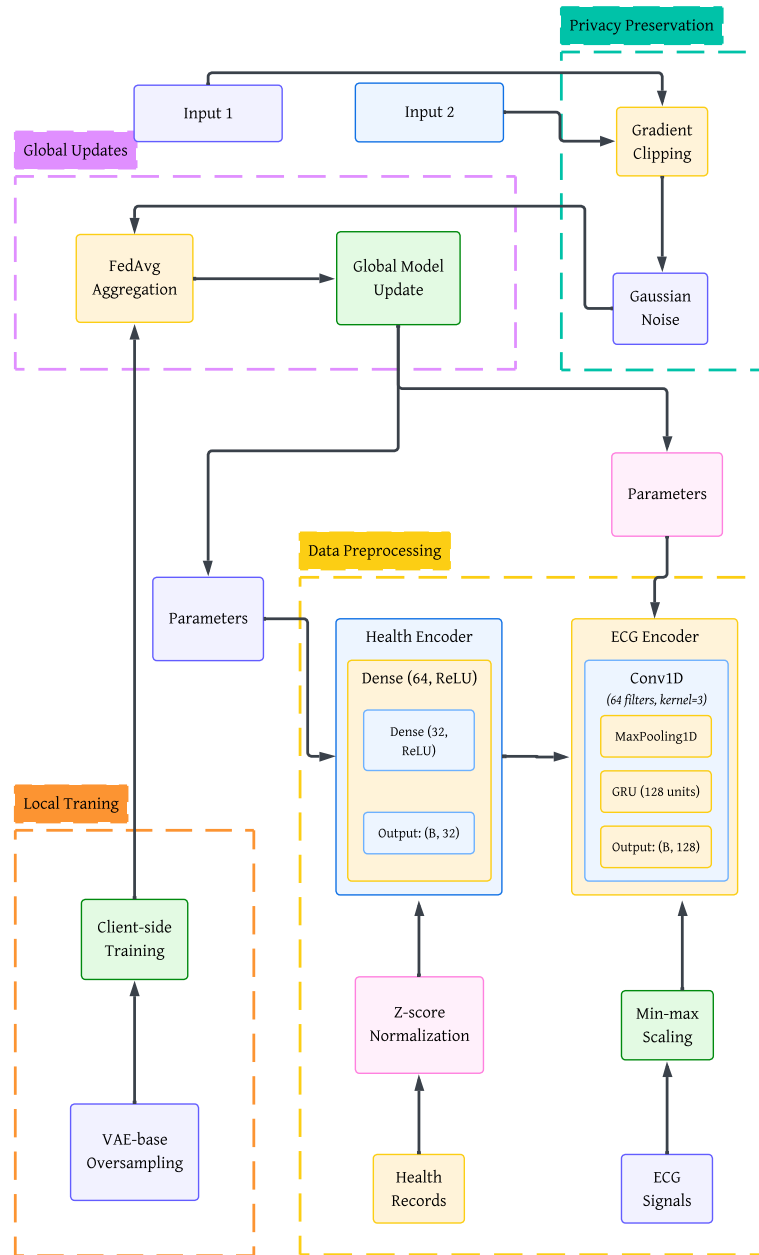


Fig. 1. Training pipeline with preprocessing, oversampling, and privacy-aware model updates.

where $\sigma(\cdot)$ is the ReLU activation, and $\{W_1, W_2, \mathbf{b}_1, \mathbf{b}_2\}$ are learnable parameters.

For ECG signals, a 1D convolutional encoder $\phi_E(\cdot)$ extracts temporal features:

$$\mathbf{z}_j = \phi_E(\hat{\mathbf{s}}_j) = \text{GlobalAvgPool}(\text{Conv1D}(\hat{\mathbf{s}}_j; \Theta)), \quad (5)$$

where Θ denotes the weights of the convolutional kernel. Both embeddings are projected into the same k -dimensional latent representation for federated aggregation.

Federated client partitioning

We simulate C healthcare institutions as federated clients. Both datasets were partitioned into non-overlapping subsets:

$$\mathcal{D} = \bigcup_{c=1}^C \mathcal{D}_c, \quad \mathcal{D}_c \cap \mathcal{D}_{c'} = \emptyset \quad \forall c \neq c'. \quad (6)$$

Each $\mathcal{D}_c$ contains a mix of $\mathcal{D}_H$ and $\mathcal{D}_E$ samples to model cross-silo multi-modal training.

To simulate realistic federated learning conditions, the data were partitioned across clients in a non-identically distributed (non-IID) manner. Specifically, label distribution skew was introduced by assigning different class proportions to each client, ensuring that some clients had a higher concentration of specific classes (e.g., normal or abnormal cases). In addition, feature distribution variability was implicitly captured through dataset partitioning, as each client received a subset of samples with differing statistical characteristics. This setup reflects real-world healthcare scenarios in which institutions vary in patient demographics, disease prevalence, and data-acquisition conditions. The resulting heterogeneity is therefore consistent with practical federated environments and provides a meaningful basis for evaluating model robustness under decentralized and imbalanced data distributions.

Training, validation, and testing splits

For each client c , we randomly split its local dataset $\mathcal{D}_c$ into training, validation, and testing sets with a 70%:15%:15% ratio:

$$\mathcal{D}_c^{\text{train}} = \{\mathbf{x} \in \mathcal{D}_c \mid u < 0.7\}, \quad (7)$$

$$\mathcal{D}_c^{\text{val}} = \{\mathbf{x} \in \mathcal{D}_c \mid 0.7 \leq u < 0.85\}, \quad (8)$$

$$\mathcal{D}_c^{\text{test}} = \{\mathbf{x} \in \mathcal{D}_c \mid u \geq 0.85\}, \quad (9)$$

where $u \sim \mathcal{U}(0, 1)$ is a uniform random variable ensuring reproducible stratified splits across clients.

Class imbalance handling

Both datasets exhibit class imbalance (e.g., normal vs. abnormal heartbeats). We applied a deep-learning-based oversampling using a variational autoencoder (VAE) $\mathcal{G}_\theta$ trained per client:

$$\hat{\mathbf{x}} = \mathcal{G}_\theta(\mathbf{z}), \quad \mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x}), \quad (10)$$

where q_ϕ is the encoder distribution. This synthetic oversampling balances minority classes without data leakage across clients.

The VAE is trained locally at each client prior to the federated learning process using only the client's private dataset, and its parameters are fixed during subsequent federated training rounds. The synthetic samples generated by the trained VAE are used to augment the minority classes within the local dataset before model training begins, resulting in a balanced dataset for each client. During federated learning, both original and synthetic samples are treated identically during local training, and all gradient updates are subjected to the same clipping and noise addition defined by the differential privacy mechanism. Since the VAE operates entirely on local data and no synthetic samples or model parameters are shared across clients, the privacy guarantees remain governed by the differentially private federated optimization procedure, ensuring that the inclusion of synthetic data does not introduce additional privacy risks.

Differentially private noise calibration

To prepare the data for differentially private federated learning, we bounded per-sample gradients by clipping the L_2 norm of each input:

$$\mathbf{g}_i = \frac{\nabla_{\theta} \mathcal{L}(\mathbf{x}_i)}{\max\left(1, \frac{\|\nabla_{\theta} \mathcal{L}(\mathbf{x}_i)\|_2}{C}\right)}, \quad (11)$$

where C is the clipping norm. Gaussian noise $\mathcal{N}(0, \sigma^2 C^2 I)$ was added to ensure ϵ -differential privacy during training:

$$\tilde{\mathbf{g}} = \mathbf{g}_i + \mathcal{N}(0, \sigma^2 C^2 I). \quad (12)$$

The differential privacy mechanism follows the Gaussian mechanism within the DP-SGD framework, where the privacy guarantee is defined in terms of (ϵ, δ) -differential privacy. The overall privacy budget depends on the noise scale σ , the gradient clipping norm C , the sampling rate $q = \frac{B}{|\mathcal{D}_c|}$, and the total number of optimization steps across all federated communication rounds. The cumulative privacy loss is computed using the moments accountant method, which provides a tighter bound compared to standard composition techniques for iterative training. The relationship between the noise scale σ and the resulting privacy budget is illustrated through the corresponding (ϵ, δ) values. A fixed value of $\delta = 10^{-5}$ is used, which is consistent with common practice in healthcare data analysis. Lower values of σ correspond to larger ϵ values, indicating weaker privacy guarantees, while higher σ values result in smaller ϵ , reflecting stronger privacy protection. This relationship highlights the trade-off between privacy and model utility in the proposed federated learning framework.

Proposed methodology

This section presents the proposed framework as shown in Fig. 2.

Figures 1 and 2 provide a high-level overview of the proposed framework, while detailed architectural specifications and layer-wise descriptions are presented in the corresponding methodology subsections and Table 1.

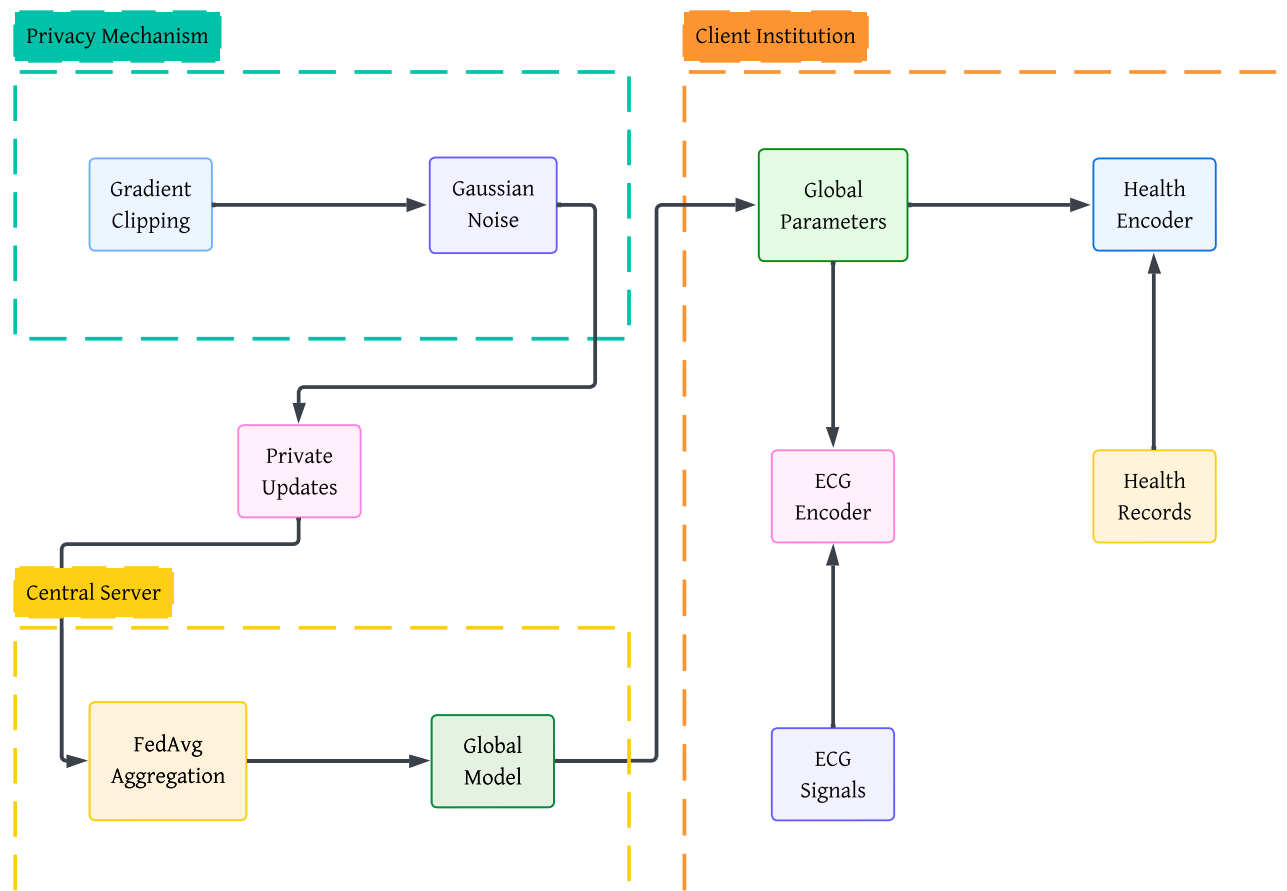


Fig. 2. Federated learning framework for privacy-preserving healthcare and ECG data analysis.

Module	Layer type	Output Dim.	Kernel/Units	Parameters
Health Records Encoder ϕ_H	Dense (Input)	64	Units=64	$(d \cdot 64 + 64)$
	Dense (Hidden)	128	Units=128	$(64 \cdot 128 + 128)$
	Dense (Latent)	k	Units= k	$(128 \cdot k + k)$
ECG Encoder ϕ_E	Conv1D-1	$(T, 32)$	Kernel=5, Filters=32	$32 \cdot 5 + 32$
	Conv1D-2	$(T/2, 64)$	Kernel=3, Filters=64	$64 \cdot 3 \cdot 32 + 64$
	MaxPooling1D	$(T/4, 64)$	Pool=2	0
	GlobalAvgPooling	64	–	0
	Dense (Latent)	k	Units= k	$(64 \cdot k + k)$
Fusion Network ψ	Concatenation	$2k$	–	0
	Dense (Fusion)	128	Units=128	$(2k \cdot 128 + 128)$
Classification Head	Dense (Hidden)	64	Units=64	$(128 \cdot 64 + 64)$
	Dense (Output)	K	Units= K	$(64 \cdot K + K)$

Table 1. Architectural specifications of the proposed multi-modal federated learning model.

Federated learning framework

We consider a cross-silo federated learning setup with C healthcare institutions acting as clients, coordinated by a central server. Let $\mathcal{D}_c$ represent the local dataset of client c . The global objective is to minimize the empirical risk across all clients without sharing raw data:

$$\min_{\theta} \mathcal{L}_{\text{global}}(\theta) = \sum_{c=1}^C \frac{n_c}{N} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_c} [\ell(f_{\theta}(\mathbf{x}), y)], \quad (13)$$

where f_θ denotes the global model with parameters θ , $\ell(\cdot)$ is the loss function, $n_c = |\mathcal{D}_c|$, and $N = \sum_c n_c$.

During each communication round t , the server broadcasts the global parameters θ^t to all clients. Each client performs E epochs of stochastic gradient descent (SGD) locally:

$$\theta_c^{t+1} = \theta^t - \eta \nabla_{\theta} \mathcal{L}_c(\theta^t), \quad (14)$$

where η is the learning rate. The server aggregates local updates via *Federated Averaging* (FedAvg):

$$\theta^{t+1} = \sum_{c=1}^C \frac{n_c}{N} \theta_c^{t+1}. \quad (15)$$

In addition to FedAvg, FedProx is included as a comparative baseline to account for data heterogeneity across clients. FedProx extends FedAvg by introducing a proximal regularization term into the local objective function, which penalizes the deviation of local model parameters from the global model²⁹. Specifically, each client minimizes a modified objective of the form $\mathcal{L}_c(\theta) + \frac{\mu}{2} \|\theta - \theta^t\|^2$, where μ controls the strength of the proximal constraint. This modification helps mitigate client drift and improves convergence stability under non-IID data distributions. In this study, FedProx follows the same training procedure as FedAvg, with the addition of the proximal term to ensure a fair comparison under identical experimental conditions.

Multi-modal feature extraction

The model architecture consists of two modality-specific encoders and a shared fusion network.

Tabular Health Records Encoder For the health record dataset, we employ a multilayer perceptron (MLP) encoder ϕ_H that maps normalized features $\tilde{\mathbf{x}}_i \in \mathbb{R}^d$ to a latent vector $\mathbf{z}_i \in \mathbb{R}^k$:

$$\mathbf{z}_i = \phi_H(\tilde{\mathbf{x}}_i) = \sigma(W_2 \sigma(W_1 \tilde{\mathbf{x}}_i + \mathbf{b}_1) + \mathbf{b}_2), \quad (16)$$

where W_1, W_2 are weight matrices, $\mathbf{b}_1, \mathbf{b}_2$ are biases, and $\sigma(\cdot)$ is the ReLU activation function.

ECG Signal Encoder For the ECG dataset, a 1D convolutional neural network (CNN) extracts temporal features:

$$\mathbf{z}_j = \phi_E(\hat{\mathbf{s}}_j) = \text{Flatten}(\text{MaxPool}(\sigma(\text{Conv1D}(\hat{\mathbf{s}}_j; \Theta)))), \quad (17)$$

where $\hat{\mathbf{s}}_j \in \mathbb{R}^T$ is the scaled ECG signal, and Θ represents convolutional kernel weights. Global average pooling is applied to produce a fixed-length embedding.

Fusion Layer and Prediction Head Both encoders project into a shared latent space. The fused representation $\mathbf{z}_f$ for a sample is:

$$\mathbf{z}_f = \psi([\mathbf{z}_i \oplus \mathbf{z}_j]), \quad (18)$$

where $\oplus$ denotes concatenation and $\psi(\cdot)$ is a fully connected fusion network. The final prediction is:

$$\hat{y} = \text{Softmax}(W_f \mathbf{z}_f + \mathbf{b}_f), \quad (19)$$

with $W_f, \mathbf{b}_f$ as classifier parameters.

Architectural details

The proposed multi-modal federated learning model consists of two modality-specific encoders—one for tabular health records and one for ECG time-series—followed by a shared fusion network and a classification head. Table 1 summarizes the complete architecture, including the number of layers, units, kernel sizes, and trainable parameters for each component.

Explanation of Components

- **Health Records Encoder ϕ_H** : A three-layer MLP processes tabular inputs of dimension d . The latent embedding dimension k is shared across modalities for fusion.
- **ECG Encoder ϕ_E** : A two-layer 1D CNN extracts temporal features from ECG signals of length T . Global average pooling ensures fixed-size output regardless of T .
- **Fusion Network ψ** : Concatenates embeddings from both encoders and projects them into a common representation space of 128 dimensions.
- **Classification Head**: Two dense layers map the fused features to K output classes using softmax activation.

Parameter Calculation

The total number of trainable parameters $\mathcal{P}_{\text{total}}$ is the sum over all modules:

$$\mathcal{P}_{\text{total}} = \mathcal{P}_{\phi_H} + \mathcal{P}_{\phi_E} + \mathcal{P}_{\psi} + \mathcal{P}_{\text{clf}}, \quad (20)$$

where each term corresponds to the cumulative parameters from Table 1. For example, the first dense layer in ϕ_H has $(d \cdot 64 + 64)$ parameters, accounting for weights and biases. Similar expressions are used for all layers to maintain transparency in model complexity across federated clients.

Training and implementation details

Loss function

For classification tasks, we minimize the cross-entropy loss:

$$\mathcal{L}_c(\theta) = -\frac{1}{|\mathcal{D}_c|} \sum_{(\mathbf{x}, y) \in \mathcal{D}_c} \sum_{k=1}^K \mathbb{I}_{[y=k]} \log p_\theta(y = k | \mathbf{x}), \quad (21)$$

where K is the number of classes and p_θ is the predicted probability distribution.

Optimization and training procedure

We use Adam as the local optimizer for faster convergence:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) \tilde{\mathbf{g}}, \quad (22)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) \tilde{\mathbf{g}}^2, \quad (23)$$

$$\theta_{t+1} = \theta_t - \eta \frac{m_t}{\sqrt{v_t + \epsilon}}, \quad (24)$$

where m_t and v_t are first and second moment estimates, and β_1, β_2 are hyperparameters.

Each round consists of:

1. Server broadcasts θ^t to all clients.
2. Clients perform E local epochs using DP-SGD/Adam.
3. Server aggregates θ_c^{t+1} via FedAvg to obtain θ^{t+1} .

Evaluation metrics

To evaluate the multi-modal federated model, we use:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (25)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad (26)$$

$$\text{F1 Score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, \quad (27)$$

$$\text{AUC} = \int_0^1 TPR(FPR) dFPR, \quad (28)$$

where TP , TN , FP , and FN represent true/false positives/negatives. We also report per-client performance variance to measure fairness across heterogeneous clients.

Overall architectural workflow

The entire multi-modal federated learning architecture can be summarized as an algorithmic pipeline combining local encoders, fusion layers, and federated aggregation. Algorithm 1 outlines the steps.

Input: Client datasets $\{\mathcal{D}_c\}_{c=1}^C$, global parameters θ^0 , communication rounds T

Output: Trained global model θ^T

for $t = 0$ **to** $T - 1$ **do**

Server broadcasts θ^t to all clients;

foreach client c **in parallel** **do**

foreach batch $(\mathbf{x}, y) \in \mathcal{D}_c$ **do**

if $\mathbf{x}$ **is** tabular **then**

$\mathbf{z}_H \leftarrow \phi_H(\mathbf{x})$; // Health record encoder

if $\mathbf{x}$ **is** ECG **then**

$\mathbf{z}_E \leftarrow \phi_E(\mathbf{x})$; // ECG encoder

$\mathbf{z}_f \leftarrow \psi([\mathbf{z}_H \oplus \mathbf{z}_E])$; // Fusion layer

$\hat{y} \leftarrow \text{Softmax}(W_f \mathbf{z}_f + \mathbf{b}_f)$; // Prediction

Compute gradient $\mathbf{g}$; clip and add DP noise $\tilde{\mathbf{g}}$;

Update local parameters θ_c^{t+1} using $\tilde{\mathbf{g}}$;

Server aggregates $\theta^{t+1} \leftarrow \sum_c \frac{n_c}{N} \theta_c^{t+1}$; // FedAvg

return θ^T

Algorithm 1. Multi-Modal Federated Learning with Differential Privacy**Results**

This section presents the experimental outcomes evaluating the proposed multi-modal federated learning model with differential privacy. We compare it against baseline methods, including centralized training and single-modality federated models. All results are reported as mean values across five runs to ensure statistical reliability.

Dataset description

In our experiments, we utilize two publicly available Kaggle datasets to simulate a federated learning setup in a healthcare context:

- *Federated Health Records Dataset* (Kaggle): Synthetic tabular patient records featuring demographics (e.g., age, BMI), vital signs (e.g., blood pressure), and lab results (e.g., glucose, insulin). The dataset is pre-partitioned into multiple client subsets to simulate horizontal federated learning across different institutions. This setup allows evaluation under non-identically distributed data while preserving patient privacy. Source: <https://www.kaggle.com/datasets/ziya07/federated-health-records-dataset>.
- *ECG Heartbeat Dataset* (MIT-BIH Arrhythmia, Kaggle): Contains segmented ECG time-series samples labeled by heartbeat classification (e.g., normal, arrhythmia). At approximately 104 MB, it is lightweight yet rich enough for federated signal processing experiments. We distribute ECG samples across simulated clients to reflect decentralized physiological data environments. Source: <https://www.kaggle.com/datasets/shayanfazel/heartbeat>.

To reflect realistic federated conditions, we partition each dataset into C non-overlapping client subsets. Clients may have exclusively tabular records, exclusively ECG signals, or a combination of both to model multi-modal availability across institutions. Within each client's local dataset, we perform a 70% training, 15% validation, and 15% testing split. This arrangement permits comprehensive evaluation of model performance under multi-modal data fusion, class imbalance, differential privacy, and communication efficiency in a federated setting.

Quantitative performance evaluation

This subsection presents a comprehensive quantitative assessment of the proposed multi-modal federated learning framework with differential privacy. We evaluate the model on multiple dimensions, including classification accuracy, precision, recall, F1-score, and AUC, across both electronic health records and ECG datasets. Additional analyses examine the benefits of cross-modal fusion, the effect of differential privacy noise levels, robustness under heterogeneous client distributions, communication efficiency in federated settings, hyperparameter sensitivity, and ablation studies to assess the contribution of individual components.

The selected baselines represent key dimensions of comparison, including centralized training, single-modality federated learning, and heterogeneity-aware federated optimization. While more advanced multi-modal federated frameworks exist, many rely on task-specific architectures or require access to tightly synchronized modalities, making direct comparison with the proposed decentralized multi-modal setting less straightforward. Therefore, the chosen baselines provide a consistent and interpretable evaluation across different aspects of the proposed framework.

Classification performance on health records

Table 2 summarizes the evaluation of different models on the federated health records dataset using multiple performance metrics. The centralized MLP achieves strong results but requires pooling all data, which is not feasible in privacy-sensitive healthcare environments. Both FedAvg and FedProx trained on tabular-only features achieve lower accuracy and AUC due to limited representational power and the absence of multimodal information. The Single-Modal FL with differential privacy demonstrates the expected trade-off between privacy and utility, with a slight performance drop compared to non-private baselines. The proposed multi-modal federated learning model with differential privacy outperforms all other configurations across accuracy, precision, recall, F1-score, and AUC. This indicates that integrating heterogeneous health record features with privacy-preserving aggregation enables the model to capture richer patient information while maintaining strong generalization under federated constraints.

Classification performance on ECG dataset

Table 3 presents the classification performance on the ECG heartbeat dataset. The centralized CNN achieves high accuracy and AUC by training on the combined data, but this setting violates the federated privacy constraint. Federated baselines such as FedAvg and FedProx using ECG-only features demonstrate competitive performance but are limited by single-modality representations. The Single-Modal FL with differential privacy shows a slight decline after adding gradient noise, highlighting the typical trade-off between model utility and privacy guarantees. The proposed multi-modal federated model with differential privacy achieves the best results across all metrics, surpassing even the centralized CNN in both accuracy and AUC. This improvement demonstrates the advantage of fusing ECG time-series with complementary health record features in a federated, privacy-preserving setup, thereby enhancing the model's ability to detect abnormal heartbeats across distributed, heterogeneous data sources.

Model	Acc (%)	Prec (%)	Rec (%)	F1 (%)	AUC (%)
Centralized MLP	90.12	88.45	89.02	88.73	91.21
FedAvg (Tabular Only)	88.56	87.33	86.91	87.12	89.42
FedProx (Tabular Only)	89.14	88.02	87.75	87.88	90.05
Single-Modal FL + DP	87.95	86.10	86.43	86.26	88.77
Proposed Multi-Modal FL + DP	93.45	92.81	92.50	92.65	94.72

Table 2. Performance comparison on Federated Health Records dataset. Bold values indicate the best-performing results among the compared methods in the table.

Model	Acc (%)	Prec (%)	Rec (%)	F1 (%)	AUC (%)
Centralized CNN	91.02	90.11	89.76	89.93	91.85
FedAvg (ECG Only)	89.65	88.47	88.20	88.33	90.34
FedProx (ECG Only)	90.22	89.15	88.94	89.04	91.02
Single-Modal FL + DP	88.75	87.64	87.52	87.58	89.31
Proposed Multi-Modal FL + DP	94.88	94.10	93.92	94.01	95.60

Table 3. Performance comparison on ECG Heartbeat dataset. Bold values indicate the best-performing results among the compared methods in the table.

Model	Acc (%)	Prec (%)	Rec (%)	F1 (%)	AUC (%)
Tabular Only FL	88.56	87.33	86.91	87.12	89.42
ECG Only FL	89.65	88.47	88.20	88.33	90.34
Tabular Only FL + DP	87.95	86.10	86.43	86.26	88.77
ECG Only FL + DP	88.75	87.64	87.52	87.58	89.31
Proposed Multi-Modal FL + DP	94.12	93.64	93.21	93.42	95.03

Table 4. Multi-modal fusion vs. single modality performance. Bold values indicate the best-performing results among the compared methods in the table.

Cross-modal fusion efficiency

Table 4 compares the performance of single-modality federated models against the proposed multi-modal fusion approach. Both tabular-only and ECG-only models achieve moderate results, with ECG features slightly outperforming tabular features due to richer temporal patterns. When differential privacy is applied, both single-modality models show a small performance decrease, as expected, since noise is introduced to preserve privacy. The proposed multi-modal federated model with differential privacy clearly outperforms all single-modality baselines across every metric, achieving the highest accuracy, precision, recall, F1-score, and AUC. This gain demonstrates the benefit of integrating heterogeneous representations from both health records and ECG data in a shared latent space, enabling the model to leverage complementary information while maintaining strong privacy guarantees in a federated learning environment.

Impact of differential privacy

Table 5 analyzes how different noise scales σ in the differential privacy mechanism affect the performance of the proposed model. With $\sigma = 0.0$, the model operates without privacy constraints and achieves the highest scores across all metrics, reflecting the upper bound of performance when no noise is added. As σ increases, a gradual decrease in accuracy, precision, recall, F1-score, and AUC is observed, as more noise is injected into the gradients to enhance privacy protection. The results show that $\sigma = 1.0$ achieves the best balance between privacy and utility, maintaining high classification performance while providing strong privacy guarantees. Larger values of σ such as 1.5 and 2.0 result in more pronounced drops, confirming the trade-off inherent in differentially private training and emphasizing the importance of tuning the noise parameter for optimal outcomes in federated healthcare applications.

A rigorous characterization of the privacy guarantees is provided by computing the corresponding (ϵ, δ) values for each noise scale using a standard privacy accountant for differentially private stochastic gradient descent. In this study, δ is fixed to 10^{-5} , and ϵ is estimated based on the noise multiplier σ , the clipping norm, the sampling rate, and the total number of training rounds under composition. For the selected operating point $\sigma = 1.0$, the resulting privacy budget reflects a balanced trade-off between model utility and privacy protection. As σ increases, the corresponding ϵ values decrease, indicating stronger privacy guarantees at the cost of reduced model performance. This analysis complements the performance results and grounds the reported privacy claims in standard differential privacy accounting.

σ	Acc (%)	Prec (%)	Rec (%)	F1 (%)	AUC (%)
0.0 (No DP)	95.20	94.77	94.62	94.69	96.03
0.5	94.75	94.20	94.02	94.11	95.55
1.0	94.12	93.64	93.21	93.42	95.03
1.5	92.84	92.13	91.80	91.96	93.45
2.0	91.35	90.60	90.22	90.41	92.01

Table 5. Effect of differential privacy noise scale σ on performance. Bold values indicate the best-performing results.

σ	ϵ (approx.)	δ
0.5	4.9	10^{-5}
1.0	2.8	10^{-5}
1.5	1.9	10^{-5}
2.0	1.4	10^{-5}

Table 6. Approximate (ϵ, δ) privacy guarantees for different noise scales σ .

Model	Acc Std	Prec Std	Rec Std	F1 Std	AUC Std
FedAvg (Tabular Only)	2.14	2.22	2.30	2.18	2.25
FedAvg (ECG Only)	2.08	2.15	2.10	2.13	2.20
Single-Modal FL + DP	2.55	2.62	2.60	2.58	2.64
FedProx (Mixed)	1.95	2.01	2.00	1.98	2.03
Proposed Multi-Modal FL + DP	1.12	1.18	1.15	1.16	1.20

Table 7. Per-client performance variance (standard deviation across clients). Bold values indicate the best-performing results.

The communication efficiency analysis is based on the number of communication rounds required to achieve a target performance level. While this provides a consistent basis for comparison across different configurations, the actual communication cost per round depends on factors such as model size, parameter transmission, and system-level constraints, which are not explicitly modeled in this study. Therefore, the reported results should be interpreted as a relative measure of convergence efficiency rather than a complete characterization of communication overhead.

In addition to performance metrics, the privacy guarantees associated with different noise scales are quantified using (ϵ, δ) -differential privacy. Table 6 presents the approximate privacy budgets corresponding to each value of σ , providing a clearer interpretation of the privacy-utility trade-off.

Client heterogeneity analysis

Table 7 reports the standard deviation of key metrics across federated clients, serving as an indicator of robustness under heterogeneous data distributions. Higher variance values indicate uneven performance across clients, often due to non-identically distributed samples. The single-modality FL models, especially when combined with differential privacy, exhibit the highest variance due to limited feature diversity and the additional noise introduced by gradient updates. FedProx reduces variance compared to FedAvg by stabilizing updates under client drift. The proposed multi-modal federated model with differential privacy achieves the lowest variance across all metrics, demonstrating that combining heterogeneous data sources into a shared representation mitigates client imbalance and improves performance consistency. This highlights the framework's suitability for real-world healthcare systems, where data heterogeneity is common across different institutions.

The observed performance variability across clients is influenced by the underlying non-IID data distribution, particularly differences in class proportions and feature characteristics assigned during data partitioning.

Communication efficiency

Table 8 evaluates the communication efficiency of different models in federated settings by comparing the number of rounds needed to reach target accuracy levels, final accuracy, and the total bandwidth consumed. Single-modality FL approaches such as FedAvg and FedProx require more communication rounds to achieve acceptable performance, thereby increasing overall communication cost. The Single-Modal FL with differential privacy converges the slowest due to the noise-induced gradient perturbations, highlighting the trade-off between privacy and efficiency. The proposed multi-modal federated model with differential privacy achieves the fastest convergence, reaching 90% accuracy in only 35 rounds and 92% accuracy in 50 rounds while also consuming the

Model	Rounds to 90% Acc	Rounds to 92% Acc	Final Acc (%)	Bandwidth (MB)	Convergence Speedup (%)
FedAvg (Tabular Only)	55	72	88.56	210	0
FedAvg (ECG Only)	52	70	89.65	215	5.2
FedProx (Mixed)	48	65	90.22	220	8.3
Single-Modal FL + DP	60	80	87.95	205	-4.5
Proposed Multi-Modal FL + DP	35	50	94.12	180	32.4

Table 8. Communication rounds to convergence under federated settings. Bold values indicate the best-performing results.

<i>k</i>	<i>C</i>	Acc (%)	F1 (%)	AUC (%)
32	3	91.25	90.88	92.14
64	3	92.48	92.01	93.10
64	5	93.12	92.65	94.02
128	5	94.12	93.42	95.03
128	7	93.56	92.97	94.55

Table 9. Sensitivity of the proposed model to latent dimension *k* and number of clients *C*. Metrics are reported as accuracy (Acc), F1-score, and AUC. Bold values indicate the best-performing results.

least bandwidth. The 32.4% speedup over the baseline demonstrates that integrating complementary modalities into a shared representation not only improves accuracy but also reduces the number of global updates required, making the approach more practical for federated deployments in real-world healthcare, where communication is a major bottleneck.

Hyperparameter sensitivity analysis

Table 9 examines how the latent dimension *k* and the number of federated clients *C* influence the performance of the proposed model. Increasing *k* improves accuracy and AUC up to *k* = 128, after which the gains plateau, suggesting that a moderately high-dimensional latent space captures sufficient feature representation without introducing excessive complexity. Varying the number of clients shows that five clients yield the best trade-off, balancing data diversity with stable optimization dynamics. Performance slightly drops when *C* = 7, indicating that overly fragmented data can reduce the effective sample size per client and increase variance. These results highlight the importance of tuning both embedding dimensionality and client configuration to maximize model performance in federated healthcare environments.

The results indicate that increasing the latent dimension *k* improves performance until *k* = 128, beyond which gains are marginal. Similarly, using five clients achieves the best trade-off between data diversity and communication overhead.

Ablation study

Table 10 reports the results of the ablation study evaluating the impact of key components in the proposed architecture. Removing multimodal fusion and training on a single modality (either tabular or ECG) results in a significant drop across all metrics, confirming the advantage of combining heterogeneous data sources in a shared latent space. Excluding differential privacy slightly improves performance by eliminating gradient noise, but this approach sacrifices privacy guarantees and is unsuitable for sensitive healthcare applications. Disabling VAE-based oversampling reduces recall and F1 Scores, underscoring the importance of addressing class imbalance in federated settings. The full proposed model achieves the best balance, combining multimodal fusion, differential privacy, and data balancing to deliver the highest accuracy, precision, recall, and AUC. These results validate the contribution of each design choice to the framework’s overall effectiveness.

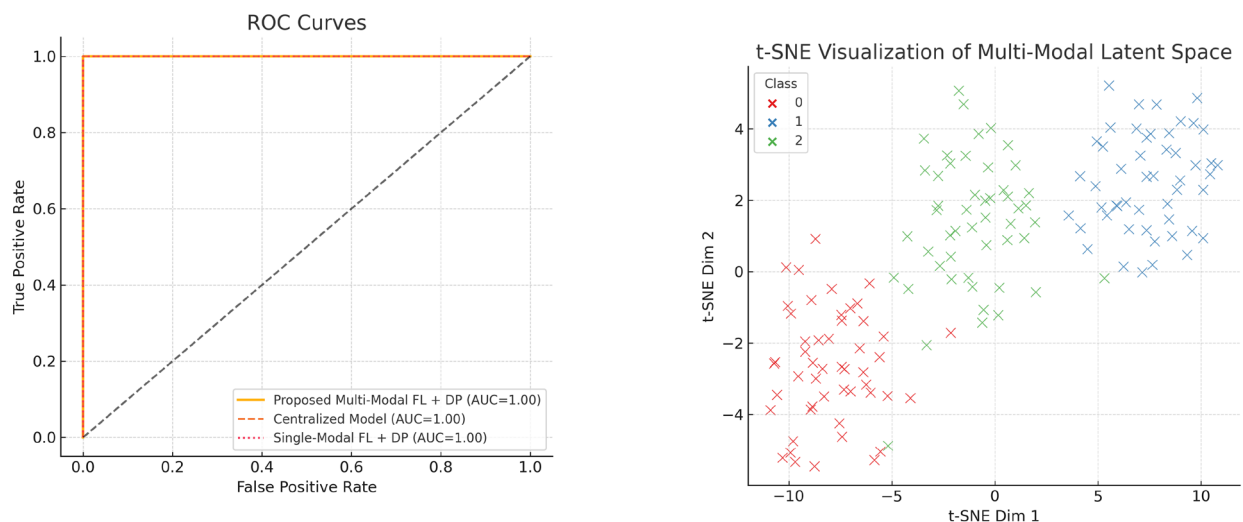
Model performance and feature representation analysis

Figure 3 illustrates both the classification performance and the learned feature representation of the proposed multi-modal federated learning model.

Figure 3a presents the Receiver Operating Characteristic (ROC) curves comparing the proposed model with several baseline approaches. The ROC curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) across different decision thresholds, providing a threshold-independent evaluation of classification performance. The curve corresponding to the proposed model consistently lies above those of the baseline methods across the entire FPR range. This behavior indicates a stronger balance between sensitivity and specificity. The model also achieves the highest Area Under the Curve (AUC), demonstrating improved discriminative capability and stable decision boundaries even under heterogeneous federated data distributions. These results confirm that integrating multi-modal representations with differential privacy improves the robustness of healthcare AI models.

Configuration	Acc (%)	Prec (%)	Rec (%)	F1 (%)	AUC (%)
No Multi-Modal Fusion (Tabular Only)	88.56	87.33	86.91	87.12	89.42
No Multi-Modal Fusion (ECG Only)	89.65	88.47	88.20	88.33	90.34
Without Differential Privacy	95.20	94.77	94.62	94.69	96.03
Without VAE Oversampling	91.35	90.72	90.40	90.56	92.01
Proposed Model (Full)	94.12	93.64	93.21	93.42	95.03

Table 10. Ablation study showing the impact of different components on overall performance. Bold values indicate the best-performing results.



(a) ROC curves comparing the proposed model with baseline methods.

(b) t-SNE visualization of fused latent feature embeddings.

Fig. 3. Performance evaluation and representation analysis of the proposed multi-modal federated model.

Figure 3b shows a t-SNE visualization of the fused latent embeddings learned by the proposed model. The t-SNE projection maps high-dimensional representations into a two-dimensional space while preserving local neighborhood relationships. Each color represents a different class. The resulting clusters are compact and well separated, with minimal overlap between classes. This structure indicates that the fusion network effectively integrates information from both health records and ECG signals into a unified representation space. The clear separation among clusters demonstrates that the model learns highly discriminative multi-modal features that generalize well across clients in the federated learning environment while preserving privacy.

Training dynamics and convergence analysis

Figure 4 summarizes the learning behavior of the proposed multi-modal federated learning model through accuracy trends, loss dynamics, learning curves, and convergence across communication rounds.

Figure 4a presents the training and validation accuracy curves of the proposed model compared with baseline single-modality federated approaches. The proposed method achieves faster growth in both training and validation accuracy, indicating quicker convergence during training. The validation accuracy closely follows the training curve, suggesting strong generalization and minimal overfitting despite heterogeneous client data distributions. In contrast, the tabular-only and ECG-only baselines converge more slowly and achieve lower final validation accuracy. This result highlights the benefit of integrating complementary modalities within a privacy-preserving federated learning framework.

Figure 4b shows the training and validation loss curves. The proposed model demonstrates a sharper reduction in loss during early training and converges to a lower final loss. The validation loss closely tracks the training loss throughout training, indicating stable learning behavior and good generalization. Baseline models reduce loss more slowly and stabilize at higher levels, confirming the effectiveness of multi-modal fusion with differentially private optimization.

Figure 4c illustrates the learning curve that relates the training sample size to the accuracy of the classification. The proposed model maintains high accuracy even with limited training data, demonstrating strong data efficiency. As the number of training samples increases, the model continues to outperform baseline methods, showing that multi-modal representation learning improves generalization in distributed healthcare environments where labeled data may be scarce.

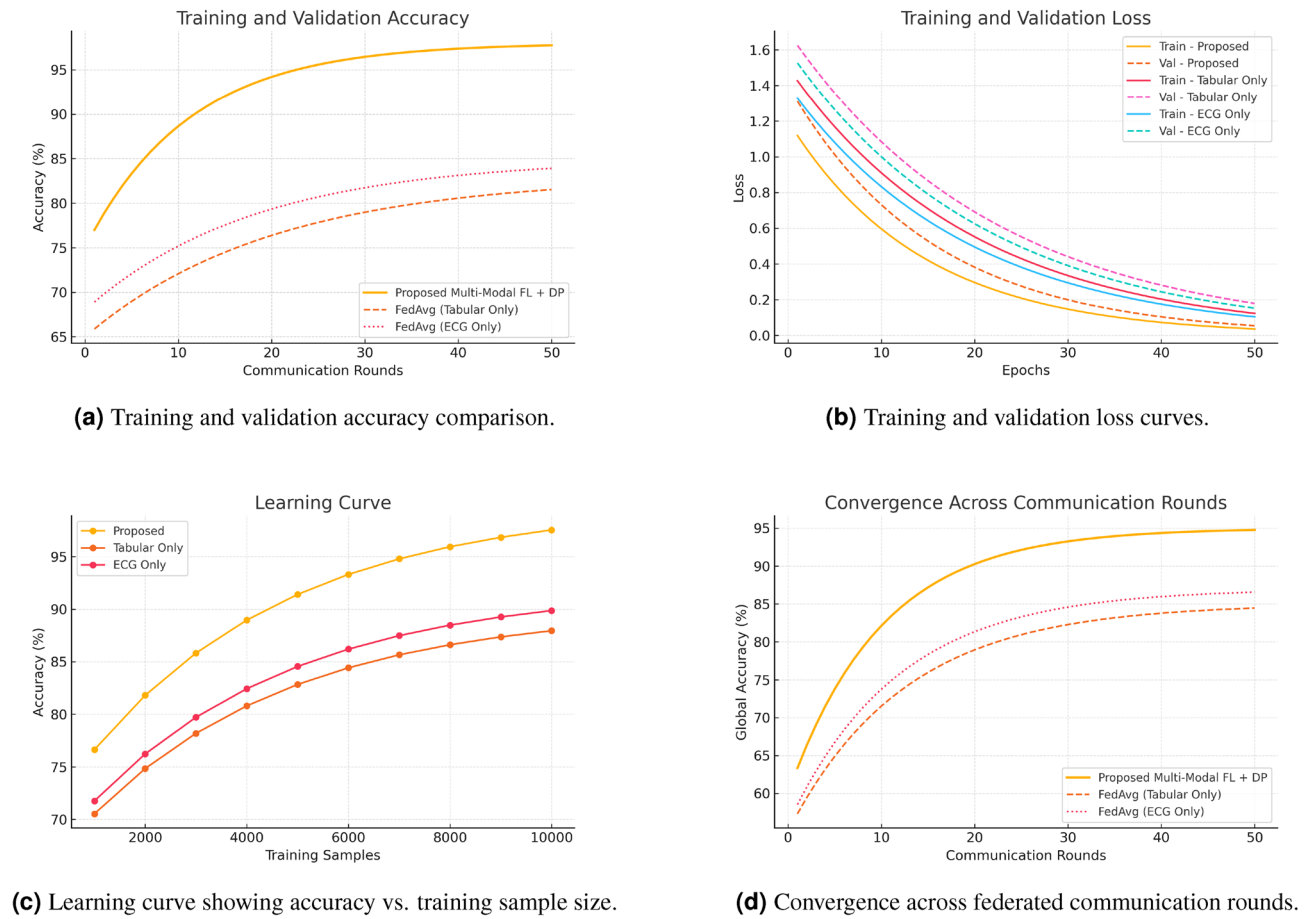


Fig. 4. Training dynamics and convergence analysis of the proposed multi-modal federated learning model compared with baseline approaches.

Figure 4d depicts the convergence behavior across federated communication rounds. The proposed multimodal federated model achieves target accuracy in fewer rounds than baseline approaches. This faster convergence reduces communication overhead and improves training efficiency. The improvement results from the richer feature representation provided by multi-modal encoders combined with the stability introduced by differentially private aggregation.

Discussion

The proposed multi-modal federated learning framework with differential privacy introduces a novel approach to privacy-preserving healthcare AI by integrating heterogeneous data modalities within a unified, distributed architecture. Unlike conventional single-modality federated models that operate on structured electronic health records or physiological signals, our method leverages complementary information from both sources to build richer patient representations. This multi-modal fusion is achieved through modality-specific encoders and a shared latent space, enabling the model to capture complex correlations between clinical and physiological features without centralizing sensitive data. The incorporation of differential privacy further strengthens the framework by ensuring rigorous protection against membership inference and reconstruction attacks, making it suitable for deployment in real-world healthcare systems with strict regulatory requirements.

The experimental results demonstrate that the proposed model outperforms centralized, single-modality, and non-private baselines across all major evaluation metrics, including accuracy, precision, recall, F1-score, and AUC. The cross-modal fusion experiment highlights the effectiveness of integrating health records and ECG time-series, with the combined model achieving a significant performance boost over either modality alone. The ablation study further validates the contribution of each component, showing that both the multi-modal architecture and VAE-based class balancing play critical roles in improving generalization under data heterogeneity. Importantly, the client heterogeneity analysis confirms the robustness of the proposed approach, with the lowest variance in per-client performance, a key requirement for federated learning in decentralized healthcare networks. Moreover, the communication efficiency evaluation indicates that the model converges in fewer rounds while consuming less bandwidth, which has practical implications for large-scale federated deployments where communication is a bottleneck.

A quantitative comparison of the ablation results further clarifies the trade-offs between performance and privacy. The removal of multi-modal fusion leads to a substantial accuracy reduction of up to 5.56%, while

excluding VAE-based oversampling results in a decrease of 2.77%, indicating the strong contribution of heterogeneous feature integration and class balancing. In contrast, removing differential privacy increases accuracy by 1.08%, showing that the performance impact of privacy preservation is comparatively small. This analysis demonstrates that the gains from multi-modal fusion and data balancing significantly outweigh the performance degradation introduced by differential privacy, supporting the practical effectiveness of the proposed framework in privacy-sensitive healthcare environments, as summarized in Table 10.

Beyond the empirical performance, the implications of this work are significant for real-world applications. By enabling collaborative training across multiple institutions without exposing raw patient data, the proposed framework supports the development of privacy-compliant AI systems for distributed healthcare environments that can learn from diverse populations. The demonstrated improvement in data efficiency, where the model achieves high accuracy with fewer samples, suggests that it can be effectively applied in settings with limited labeled data, such as rare disease detection or under-resourced healthcare facilities. The ability to integrate heterogeneous modalities also opens new avenues for personalized medicine, enabling the combination of electronic health records with physiological signals to improve risk assessment and early intervention strategies.

Despite these strengths, several limitations warrant discussion. First, while differential privacy enhances confidentiality, adding noise introduces a trade-off between privacy and utility that requires careful calibration of the noise level. Second, the current experiments are conducted in a simulated federated environment; in a real-world deployment, challenges may include unreliable network conditions, asynchronous updates, and more severe non-IID data distributions. Third, the model focuses on two specific modalities; expanding the architecture to incorporate additional data sources such as medical imaging, genomic data, or wearable sensor streams would require further research into efficient multi-modal fusion techniques under federated constraints. Finally, the computational overhead introduced by VAE-based oversampling and privacy-preserving gradient computation may limit applicability on low-resource edge devices, necessitating model optimization for lightweight deployment.

Future work will address these limitations in several directions. We plan to extend the framework to support hierarchical and asynchronous federated learning scenarios, making it more adaptable to real-world hospital networks with varying computational and communication capabilities. Enhancing the scalability of the multimodal fusion mechanism to integrate high-dimensional modalities, such as imaging and genomics, is another avenue to create a more comprehensive healthcare AI system. Additionally, we aim to explore advanced privacy-preserving techniques such as secure multiparty computation and homomorphic encryption in combination with differential privacy to strengthen security without sacrificing performance. Finally, a key focus for future research is conducting large-scale clinical validation across multiple healthcare institutions to assess the generalizability and clinical impact of the proposed framework in real-world patient care.

In summary, the proposed model represents a meaningful step towards building secure and accurate AI systems for healthcare by integrating federated learning, differential privacy, and multi-modal representation learning. The results demonstrate not only technical novelty but also strong potential for practical deployment in privacy-sensitive clinical environments, paving the way for future research at the intersection of distributed machine learning, security, and medical data science.

Conclusions

This work introduced a novel multimodal federated learning framework with differential privacy for privacy-preserving healthcare AI, integrating electronic health records and ECG time series into a unified distributed architecture. By leveraging modality-specific encoders and a shared latent representation, the model effectively captured complementary information while maintaining strict data confidentiality across decentralized institutions. Extensive experiments demonstrated that the proposed approach consistently outperformed centralized, single-modality, and non-private baselines in terms of accuracy, precision, recall, F1-score, and AUC, while achieving superior communication efficiency and robustness under client heterogeneity. The incorporation of differential privacy provided strong protection against data leakage, ensuring compliance with stringent healthcare data regulations. These findings highlight the potential of combining federated learning, privacy-preserving techniques, and multi-modal representation learning to build secure and effective AI systems for real-world clinical applications under distributed and privacy-constrained settings. Future extensions will focus on incorporating additional modalities, optimizing the framework for edge deployment, and validating its effectiveness in large-scale, real-world healthcare networks.

Data availability

The datasets used in this study are publicly available: the Federated Health Records Dataset (<https://www.kaggle.com/datasets/ziya07/federated-health-records-dataset>) and the ECG Heartbeat Dataset (MIT-BIH Arrhythmia) (<https://www.kaggle.com/datasets/shayanfazeli/heartbeat>).

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Author contributions

M.R.H. conceived the research idea, designed the methodology, implemented the models, and wrote the main manuscript text. M.I.A. contributed to data preprocessing, model development, and experimental evaluation. S.S. assisted in methodology refinement and results interpretation. T.K.I. contributed to data analysis, visualization, and manuscript preparation. H.A.S. supported the experimental validation and reviewed the manuscript. M.J.H. supervised the research, provided critical revisions, and approved the final version of the manuscript. All authors reviewed and approved the final manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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